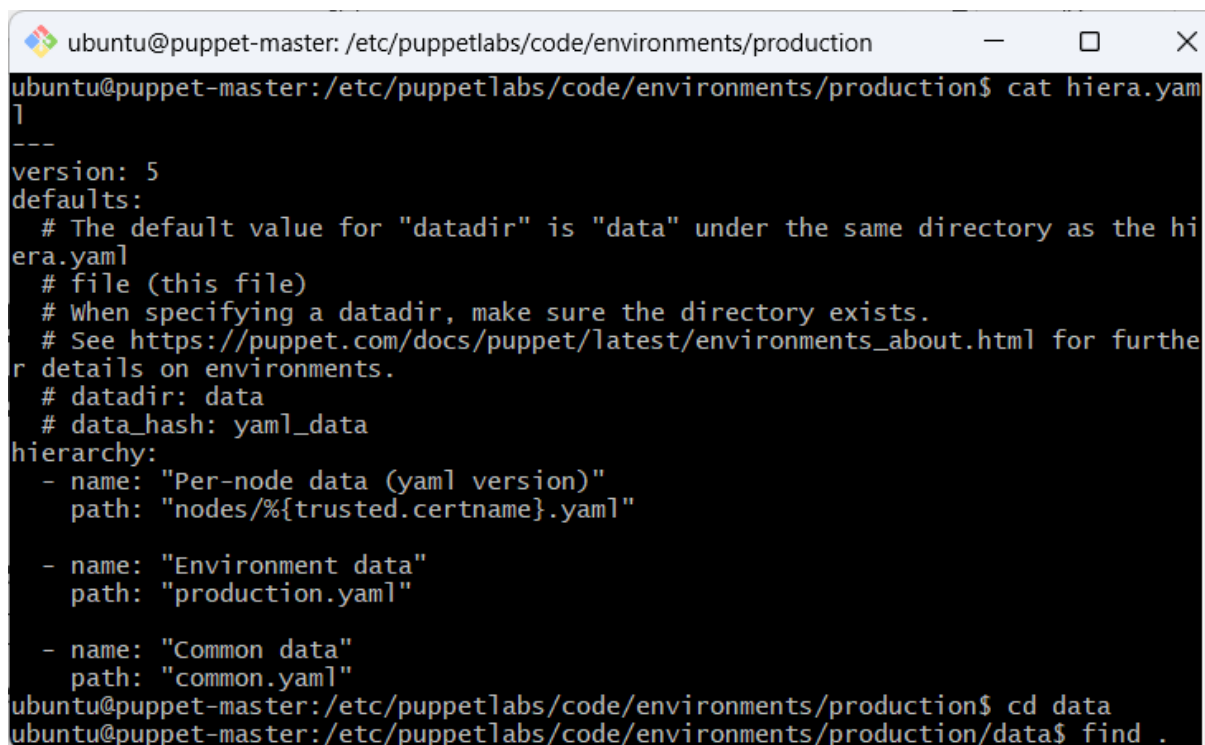


Complete Puppet Mastery Part-5 (Flagship Standalone Practical -3)

Practical 14- Enterprise Data-Driven Configuration Management Using Hiera, Automatic Parameter Lookup (APL), and Multi-Environment Puppet

Master:



```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production$ cat hiera.yaml
---
version: 5
defaults:
  # The default value for "datadir" is "data" under the same directory as the hiera.yaml
  # file (this file)
  # When specifying a datadir, make sure the directory exists.
  # See https://puppet.com/docs/puppet/latest/environments_about.html for further details on environments.
  # datadir: data
  # data_hash: yaml_data
hierarchy:
  - name: "Per-node data (yaml version)"
    path: "nodes/%{trusted.certname}.yaml"

  - name: "Environment data"
    path: "production.yaml"

  - name: "Common data"
    path: "common.yaml"
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production$ cd data
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/data$ find .
```

```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/data$ find .
./nodes
./nodes/puppet-agent01.yaml
./production.yaml
./common.yaml
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/data$ cat nodes/puppet-agent01.yaml
dynamicinfra::server_role: "Web Server"
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/data$ cat production.yaml
dynamicinfra::environment_name: "Production"
dynamicinfra::output_directory: "/opt/company"
dynamicinfra::packages:
  - curl
  - tree
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/data$ cat common.yaml
company_name: "OpenAI"
department: "Devops"
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/data$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production$ sudo nano /modules/dynamicinfra/manifests/init.pp
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production$ sudo nano mod

```

```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production$ sudo cat modules/dynamicinfra/manifests/init.pp
class dynamicinfra(
  String $server_role,
  String $environment_name,
  String $output_directory,
  Array[String] $packages
){
  class{'dynamicinfra::install':
    packages=>$packages
  }

  class{'dynamicinfra::config':
    server_role=>$server_role,
    environment_name=>$environment_name,
    output_directory=>$output_directory
  }

  class{'dynamicinfra::service': }

  Class['dynamicinfra::install']->Class['dynamicinfra::config']
  ->Class['dynamicinfra::service']
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production$ sudo cat modu

```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production$ sudo cat modules/dynamicinfra/manifests/config.pp
class dynamicinfra::config(
  String $server_role,
  String $environment_name,
  String $output_directory
){
  $service_name = $facts['os']['family'] ? {
    'Debian'=>'apache2',
    'RedHat'=>'httpd',
    default=>'apache2'
  }

  file{$output_directory:
    ensure=>directory,
    owner=>'root',
    group=>'root',
    mode=>'0755'
  }

  $company_name = lookup('company_name')
  $department = lookup('department')
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production
file{"${output_directory}/system-info.txt":
  ensure=>file,
  owner=>'root',
  group=>'root',
  mode=>'0644',

  content=>epp(
    'dynamicinfra/system-info.epp',
    {
      'server_role'=>$server_role,
      'environment_name'=>$environment_name,
      'company_name' =>$company_name,
      'department' =>$department
    }
  ),
  notify=>Service[$service_name]
}

if $facts['os']['family'] == 'Debian' {
  file{"${output_directory}/os-info.txt":
    ensure=>file,
    owner=>'root',
    group=>'root',
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production
mode=>'0644',
content=>"This node belongs to the Debian family.
Package Manager: APT\n"
}
}
elseif $facts['os']['family'] == 'RedHat' {
file{"${output_directory}/os-info.txt":
ensure=>file,
owner=>'root',
group=>'root',
mode=>'0644',
content=>"This node belongs to the Redhat family.\nPackage Manager:
DNF/YUM\n"
}
}
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production$ sudo cat modu
les/dynamicinfra/manifests/install.pp
class dynamicinfra::install
( Array[String] $packages){

package{$packages:
ensure=>installed
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production
}
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production$ sudo cat modu
les/dynamicinfra/manifests/install.pp
class dynamicinfra::install
( Array[String] $packages){

package{$packages:
ensure=>installed
}

if $facts['os']['family'] == 'Debian' {
package{'apache2':
ensure=>installed
}
}
elseif $facts['os']['family'] == 'RedHat'{
package{'httpd':
ensure=>installed
}
}
}
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production$
```

Puppet-agent01(Production Environment):

```
ubuntu@puppet-agent01: ~  
ubuntu@puppet-agent01:~$ sudo /opt/puppetlabs/bin/puppet agent -t  
Info: Refreshing CA certificate  
Info: CA certificate is unmodified, using existing CA certificate  
Info: Using environment 'production'  
Info: Retrieving pluginfacts  
Info: Retrieving plugin  
Notice: Requesting catalog from puppet-master:8140 (172.31.14.115)  
Notice: Catalog compiled by puppet-master  
Info: Caching catalog for puppet-agent01  
Info: Applying configuration version '1786021551'  
Notice: /Stage[main]/Dynamicinfra::Config/File[/opt/company/system-info.txt]/content:  
--- /opt/company/system-info.txt          2026-08-05 14:41:39.490455582 +0000  
+++ /tmp/puppet-file20260806-2270-7cmg47  2026-08-06 13:05:53.333592516 +0000  
@@ -2,6 +2,8 @@  
      Enterprise Infrastructure Report  
      -----  
  
+Company: OpenAI  
+Department: Devops  
  Server Role: Web Server  
  Environment: Production
```

```
ubuntu@puppet-agent01: ~  
  
+Company: OpenAI  
+Department: Devops  
  Server Role: Web Server  
  Environment: Production  
  
@@ -17,7 +19,7 @@  
  Architecture: amd64  
  
  Processors: 2  
-Memory: 914.00 MiB  
+Memory: 914.01 MiB  
  
  Puppet Version: 8.10.0  
  
Notice: /Stage[main]/Dynamicinfra::Config/File[/opt/company/system-info.txt]/content: content changed '{sha256}b65c0566132bb39034d5ad23cbc96e621a095be7e0c2553401cb51477da5a086' to '{sha256}bf9f961ed1c7127e38d797f9bc4255e6879734ab4625b7c6c4da1b72afe5c343'  
Info: /Stage[main]/Dynamicinfra::Config/File[/opt/company/system-info.txt]: Scheduling refresh of Service[apache2]  
Notice: /Stage[main]/Dynamicinfra::Service/Service[apache2]: Triggered 'refresh' from 1 event
```

```
ubuntu@puppet-agent01: ~  
Notice: Applied catalog in 0.45 seconds  
ubuntu@puppet-agent01:~$ ls -l /opt/company  
total 8  
-rw-r--r-- 1 root root 68 Aug 4 12:09 os-info.txt  
-rw-r--r-- 1 root root 583 Aug 6 13:05 system-info.txt  
ubuntu@puppet-agent01:~$ cat /opt/company/system-info.txt  
-----  
Enterprise Infrastructure Report  
-----  
  
Company: OpenAI  
Department: Devops  
Server Role: Web Server  
Environment: Production  
  
Hostname: puppet-agent01  
FQDN: puppet-agent01.ap-south-1.compute.internal  
Primary IP: 172.31.7.84  
  
Operating System: Ubuntu  
OS Release: 22.04  
OS Family: Debian  
  
Kernel: Linux
```

```
ubuntu@puppet-agent01: ~  
Processors: 2  
Memory: 914.01 MiB  
  
Puppet Version: 8.10.0  
  
Generated By: Puppet Master and Vishnu's Passion  
  
-----  
ubuntu@puppet-agent01:~$ sudo /opt/puppetlabs/bin/puppet agent -t  
Info: Refreshing CA certificate  
Info: CA certificate is unmodified, using existing CA certificate  
Info: Refreshing CRL  
Info: CRL is unmodified, using existing CRL  
Info: Using environment 'production'  
Info: Retrieving pluginfacts  
Info: Retrieving plugin  
Notice: Requesting catalog from puppet-master:8140 (172.31.14.115)  
Notice: Catalog compiled by puppet-master  
Info: Caching catalog for puppet-agent01  
Info: Applying configuration version '1786022629'  
Notice: Applied catalog in 0.45 seconds
```

Master configs for Agent-2:

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/development
GNU nano 6.2 hiera.yaml *
---
version: 5
defaults:
  # The default value for "datadir" is "data" under the same directory as the h
  # file (this file)
  # When specifying a datadir, make sure the directory exists.
  # See https://puppet.com/docs/puppet/latest/environments_about.html for furth
  # datadir: data
  # data_hash: yaml_data
hierarchy:
  - name: "Per-node data (yaml version)"
    path: "nodes/%{trusted.certname}.yaml"

  - name: "Environment data"
    path: "development.yaml"

  - name: "Other YAML hierarchy levels"
    path: "common.yaml"

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify  ^_ Go To Line
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/development
GNU nano 6.2 data/common.yaml
company_name: "OpenAI-DEV"
department: "development"
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/development
GNU nano 6.2 data/development.yaml
dynamicinfra:environment_name: "Development"
dynamicinfra:output_directory: "/opt/company"
dynamicinfra:experimental_features: true
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/development
GNU nano 6.2 data/nodes/puppet-agent02.yaml
dynamicinfra::server_role: "Development Server"
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/development
ubuntu@puppet-master:/etc/puppetlabs/code/environments/development$ sudo cat modules/dynamicinfra/manifests/init.pp
class dynamicinfra(
  String $server_role,
  String $environment_name,
  String $output_directory,
  Boolean $experimental_features
){
  class{'dynamicinfra::install': }

  class{'dynamicinfra::config':
    server_role=>$server_role,
    environment_name=>$environment_name,
    output_directory=>$output_directory,
    experimental_features=>$experimental_features
  }

  class{'dynamicinfra::service': }

  Class['dynamicinfra::install']->Class['dynamicinfra::config']
  ->Class['dynamicinfra::service']
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/development$ sudo cat modules/dynamicinfra/manifests/config.pp
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/development
class dynamicinfra::config(
  String $server_role,
  String $environment_name,
  String $output_directory,
  Boolean $experimental_features
){
  $service_name = $facts['os']['family'] ? {
    'Debian'=>'apache2',
    'RedHat'=>'httpd',
    default=>'apache2'
  }

  file{$output_directory:
    ensure=>directory,
    owner=>'root',
    group=>'root',
    mode=>'0755'
  }

  $company_name = lookup('company_name')
  $department = lookup('department')

  file{"${output_directory}/system-info.txt":
```



```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/development
file{"${output_directory}/system-info.txt":
  ensure=>file,
  owner=>'root',
  group=>'root',
  mode=>'0644',

  content=>epp(
    'dynamicinfra/system-info.epp',
    {
      'server_role'=>$server_role,
      'environment_name'=>$environment_name
    }
  ),
  notify=>Service[$service_name]
}

if $experimental_features{
  file { "${output_directory}/development-warning.txt":

    ensure => file,
    owner   => 'root',
    group   => 'root',
    mode     => '0644',
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/development
content => epp(
  'dynamicinfra/development-warning.epp',
  {
    'company_name'      => $company_name,
    'department'        => $department,
    'server_role'       => $server_role,
    'environment_name'  => $environment_name,
  }
),
notify => Service[$service_name],
}

}

if $facts['os']['family'] == 'Debian' {
  file{"${output_directory}/os-info.txt":
    ensure=>file,
    owner=>'root',
    group=>'root',
    mode=>'0644',
    content=>"This node belongs to the Debian family.
    Package Manager: APT\n"
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/development
}
elseif $facts['os']['family'] == 'RedHat' {
    file{"${output_directory}/os-info.txt":
        ensure=>file,
        owner=>'root',
        group=>'root',
        mode=>'0644',
        content=>"This node belongs to the Redhat family.\nPackage Manager:
        DNF/YUM\n"
    }
}
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/development$ sudo cat modules/dynamicinfra/templates/development-warning.epp
=====
DEVELOPMENT ENVIRONMENT REPORT
=====

WARNING

This node belongs to the Development Puppet Environment.

Changes deployed here are intended for testing before
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/development

Company Name : <%= $company_name %>
Department : <%= $department %>
Server Role : <%= $server_role %>
Environment : <%= $environment_name %>
Hostname : <%= $facts['networking']['hostname'] %>
Primary IP : <%= $facts['networking']['ip'] %>
Operating System : <%= $facts['os']['name'] %>
Generated By : Puppet Development Environment and Vishnu's Passion

=====
ubuntu@puppet-master:/etc/puppetlabs/code/environments/development$ sudo cat manifests/site.pp
node 'puppet-agent02'{
    include dynamicinfra
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/development$
```

Puppet-agent02(Development Environement):

```
ubuntu@puppet-agent02: ~  
ubuntu@puppet-agent02:~$ sudo /opt/puppetlabs/bin/puppet agent -t  
Info: Refreshing CA certificate  
Info: CA certificate is unmodified, using existing CA certificate  
Info: Refreshing CRL  
Info: CRL is unmodified, using existing CRL  
Info: Using environment 'development'  
Info: Retrieving pluginfacts  
Info: Retrieving plugin  
Notice: Requesting catalog from puppet-master:8140 (172.31.14.115)  
Notice: Catalog compiled by puppet-master  
Info: Caching catalog for puppet-agent02  
Info: Applying configuration version '1786109752'  
Notice: /Stage[main]/Dynamicinfra::Config/File[/opt/company/system-info.txt]/content:  
--- /opt/company/system-info.txt          2026-08-05 13:13:23.346607969 +0000  
+++ /tmp/puppet-file20260807-1427-yc9bpk  2026-08-07 13:35:53.636710695 +0000  
@@ -2,7 +2,7 @@  
-----  
Enterprise Infrastructure Report  
-----  
  
-Server Role: Application server  
+Server Role: Development Server  
Environment: Development
```

```
ubuntu@puppet-agent02: ~  
Hostname: puppet-agent02  
@@ -17,7 +17,7 @@  
Architecture: amd64  
  
Processors: 2  
-Memory: 914.01 MiB  
+Memory: 914.00 MiB  
  
Puppet Version: 8.10.0  
  
Notice: /Stage[main]/Dynamicinfra::Config/File[/opt/company/system-info.txt]/content: content changed '{sha256}509aa93d33f1605502ccc4a88963a5932fd2e81ff71a3b1cc2ec6ceab2f4bb97' to '{sha256}14341e689bbbe6151a3497313f7cda1d14b31a46873a772e28866e131484346c'  
Info: /Stage[main]/Dynamicinfra::Config/File[/opt/company/system-info.txt]: Scheduling refresh of Service[apache2]  
Notice: /Stage[main]/Dynamicinfra::Config/File[/opt/company/development-warning.txt]/content:  
--- /opt/company/development-warning.txt  2026-08-05 14:31:47.234114007 +0000  
+++ /tmp/puppet-file20260807-1427-rv5hxq  2026-08-07 13:35:53.654710890 +0000  
@@ -9,7 +9,11 @@
```

```
ubuntu@puppet-agent02: ~  
@@ -9,7 +9,11 @@  
Changes deployed here are intended for testing before  
promotion to Production.  
  
-Server Role : Application server  
+Company Name : OpenAI-DEV  
+  
+Department : development  
+  
+Server Role : Development Server  
  
Environment : Development  
  
Notice: /Stage[main]/Dynamicinfra::Config/File[/opt/company/development-warning.  
txt]/content: content changed '{sha256}da73f6188783c7bc5063881faea40f7aad08aa433  
478a0dc95e7e3612b3e1348' to '{sha256}4464bfae103d0c9443343df607d86be16e8d3cd2b21  
9cfeddb14c36f7c163f3c'  
Info: /Stage[main]/Dynamicinfra::Config/File[/opt/company/development-warning.tx  
t]: Scheduling refresh of Service[apache2]  
Notice: /Stage[main]/Dynamicinfra::Service/Service[apache2]: Triggered 'refresh'  
from 2 events  
Notice: Applied catalog in 0.58 seconds  
ubuntu@puppet-agent02:~$ ls /opt/company
```

```
ubuntu@puppet-agent02: ~  
ubuntu@puppet-agent02:~$ ls /opt/company  
development-warning.txt  os-info.txt  system-info.txt  
ubuntu@puppet-agent02:~$ sudo cat development-warning.txt  
cat: development-warning.txt: No such file or directory  
ubuntu@puppet-agent02:~$ sudo cat /opt/company/development-warning.txt  
=====
```

```
DEVELOPMENT ENVIRONMENT REPORT  
=====
```

```
WARNING  
  
This node belongs to the Development Puppet Environment.  
  
Changes deployed here are intended for testing before  
promotion to Production.  
  
Company Name : OpenAI-DEV  
  
Department : development  
  
Server Role : Development Server  
  
Environment : Development
```

```
ubuntu@puppet-agent02: ~  
Changes deployed here are intended for testing before  
promotion to Production.  
  
Company Name : OpenAI-DEV  
Department : development  
Server Role : Development Server  
Environment : Development  
Hostname : puppet-agent02  
Primary IP : 172.31.14.217  
Operating System : Ubuntu  
Generated By : Puppet Development Environment and Vishnu's Passion  
  
=====
```

```
ubuntu@puppet-agent02:~$ sudo cat /opt/company/os-info.txt  
This node belongs to the Debian family.  
    Package Manager: APT  
ubuntu@puppet-agent02:~$ |
```